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atharva@atharva-VMware-Vir  X  +  v  -  □  X

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

atharva@atharva-VMware-Virtual-Platform:~$ mkdir -p ~/practical_exam/scripts
atharva@atharva-VMware-Virtual-Platform:~$ mkdir -p ~/practical_exam/outputs
atharva@atharva-VMware-Virtual-Platform:~$ mkdir -p ~/practical_exam/logs
atharva@atharva-VMware-Virtual-Platform:~$ ls -R ~/practical_exam
/home/atharva/practical_exam:
logs  outputs  scripts

/home/atharva/practical_exam/logs:

/home/atharva/practical_exam/outputs:

/home/atharva/practical_exam/scripts:
atharva@atharva-VMware-Virtual-Platform:~$ cd ~/practical_exam
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ pwd
whoami
hostname
/home/atharva/practical_exam
atharva
atharva-VMware-Virtual-Platform
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ cat outputs/system_info.txt
cat: outputs/system_info.txt: No such file or directory
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ |
```

```
atharva@atharva-VMware-Virtual-Platform:~/practical_exam/scripts$ touch scripts/file1.txt
touch: cannot touch 'scripts/file1.txt': No such file or directory
atharva@atharva-VMware-Virtual-Platform:~/practical_exam/scripts$ cd
atharva@atharva-VMware-Virtual-Platform:~$ cd ~/practical_exam
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ touch scripts/file1.txt
touch scripts/file2.txt
touch scripts/file3.txt
touch scripts/file4.txt
touch scripts/file5.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls
logs  outputs  scripts
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls scripts
file1.txt file2.txt file3.txt file4.txt file5.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ cp scripts/file1.txt outputs/
cp scripts/file2.txt outputs/
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls outputs
file1.txt file2.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ mv scripts/file3.txt scripts/renamed_file3.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls scripts
file1.txt file2.txt file4.txt file5.txt renamed_file3.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ rm scripts/file5.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls scripts
file1.txt file2.txt file4.txt renamed_file3.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls -la
total 20
drwxrwxr-x  5 atharva atharva 4096 Jun 10 12:38 .
drwxr-x--- 18 atharva atharva 4096 Jun 10 12:37 ..
drwxrwxr-x  2 atharva atharva 4096 Jun 10 12:38 logs
drwxrwxr-x  2 atharva atharva 4096 Jun 10 12:45 outputs
drwxrwxr-x  2 atharva atharva 4096 Jun 10 12:46 scripts
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            345M  2.6M  342M   1% /run
/dev/sda2        20G   9.2G   9.4G  50% /
tmpfs            861M   0  861M   0% /dev/shm
none             1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
none             1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
```

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none          1.0M    0    1.0M    0% /run/credentials/systemd-resolved.service
tmpfs         861M   8.0K   861M    1% /tmp
tmpfs         173M   80K   172M    1% /run/user/1000
/dev/sr0      108M   108M    0 100% /run/media/atharva/CDROM
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ df -h > outputs/disk_usage.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.7Gi         944Mi        162Mi         6.3Mi         790Mi         776Mi
Swap:          2.0Gi         273Mi         1.7Gi
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ free -h > outputs/memory_usage.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  1.5  0.9 26432 15860 ?        Ss   12:08   0:37 /usr/lib/systemd/systemd --switched-root --system --deserialize=51 auto automatic
root           2  0.0  0.0    0     0 ?        S    12:08   0:00 [kthreadd]
root           3  0.0  0.0    0     0 ?        S    12:08   0:00 [pool_workqueue_release]
root           4  0.0  0.0    0     0 ?        I<   12:08   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0    0     0 ?        I<   12:08   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0    0     0 ?        I<   12:08   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0    0     0 ?        I<   12:08   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0    0     0 ?        I<   12:08   0:00 [kworker/R-netns]
root           9  0.3  0.0    0     0 ?        I    12:08   0:08 [kworker/0:0-events]
root          12  0.0  0.0    0     0 ?        I    12:08   0:00 [kworker/u512:0-ipv6_addrconf]
root          13  0.0  0.0    0     0 ?        I<   12:08   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0    0     0 ?        S    12:08   0:00 [ksoftirqd/0]
root          15  0.1  0.0    0     0 ?        I    12:08   0:04 [rcu_preempt]
root          16  0.0  0.0    0     0 ?        S    12:08   0:00 [rcu_exp_par_gp_kthread_worker/1]
root          17  0.0  0.0    0     0 ?        S    12:08   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0    0     0 ?        S    12:08   0:00 [migration/0]
root          19  0.0  0.0    0     0 ?        S    12:08   0:00 [kprobe-optimizer]
root          20  0.0  0.0    0     0 ?        S    12:08   0:00 [idle_inject/0]
root          21  0.0  0.0    0     0 ?        S    12:08   0:00 [cpuhp/0]
root          22  0.0  0.0    0     0 ?        S    12:08   0:00 [cpuhp/1]
root          23  0.0  0.0    0     0 ?        S    12:08   0:00 [idle_inject/1]
root          24  0.0  0.0    0     0 ?        S    12:08   0:00 [migration/1]
root          25  0.0  0.0    0     0 ?        S    12:08   0:00 [ksoftirqd/1]
root          28  0.0  0.0    0     0 ?        S    12:08   0:00 [cpuhp/2]
root          29  0.0  0.0    0     0 ?        S    12:08   0:00 [idle_inject/2]

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atharva@atharva-VMware-Vi X + | v
atharva 4249 0.1 0.4 239944 7668 ? Ssl 12:09 0:02 /usr/libexec/ptyxis-agent --socket-fd=3 --rlimit-nofile=1024
atharva 4322 0.0 0.2 19976 4904 pts/0 Ss+ 12:09 0:00 /usr/bin/bash
root 4360 0.0 0.0 0 0 ? I< 12:09 0:00 [kworker/u518:1-ttm]
root 4361 0.0 0.0 0 0 ? I< 12:09 0:00 [kworker/0:2H-kblockd]
root 4526 0.0 0.0 0 0 ? I 12:14 0:00 [kworker/u514:1-events_power_efficient]
root 4633 0.0 0.0 0 0 ? I 12:14 0:00 [kworker/u513:0-events_unbound]
root 4981 0.0 0.0 0 0 ? I< 12:16 0:00 [kworker/1:2H]
root 5388 0.3 2.3 580412 41712 ? Ssl 12:17 0:06 /usr/libexec/fwupd/fwupd
root 5723 0.0 0.0 0 0 ? I 12:17 0:00 [kworker/u515:2-flush-8:0]
root 5816 0.0 0.0 0 0 ? I 12:20 0:00 [kworker/u513:1-flush-8:0]
root 6334 0.3 0.0 0 0 ? I 12:22 0:05 [kworker/3:1-mpt_poll_0]
root 6529 0.0 0.4 10740 7692 ? Ss 12:22 0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root 6534 0.4 0.0 0 0 ? I 12:23 0:06 [kworker/2:0-events]
root 6542 0.0 0.0 0 0 ? I< 12:23 0:00 [kworker/u520:2]
root 7264 0.0 0.0 0 0 ? I 12:24 0:00 [kworker/u516:3-events_unbound]
root 7314 0.1 0.7 40748 12932 ? Ss 12:25 0:02 /usr/sbin/cupsd -l
cups-br+ 7315 0.0 1.1 207532 20880 ? Ssl 12:25 0:00 /usr/sbin/cups-browsed
root 8371 0.0 0.0 0 0 ? S 12:26 0:00 [psimon]
root 8946 0.0 0.0 0 0 ? I< 12:26 0:00 [kworker/0:1H]
root 9028 0.2 0.0 0 0 ? I 12:30 0:02 [kworker/3:2-mm_percpu_wq]
root 9046 0.0 0.7 18844 12896 ? Ss 12:32 0:00 sshd-session: atharva [priv]
root 9053 0.0 0.0 0 0 ? I 12:33 0:00 [kworker/u514:0-events_power_efficient]
atharva 9112 0.7 0.4 19004 7868 ? S 12:33 0:05 sshd-session: atharva@pts/1
atharva 9113 0.1 0.3 19976 5768 pts/1 Ss 12:33 0:01 -bash
root 9211 0.3 0.0 0 0 ? I 12:36 0:02 [kworker/0:3-events]
root 9212 0.5 0.0 0 0 ? I 12:37 0:03 [kworker/1:3-mm_percpu_wq]
root 9282 0.3 0.0 0 0 ? I 12:38 0:01 [kworker/2:2-events]
root 9328 0.0 0.0 0 0 ? I 12:40 0:00 [kworker/u515:1-flush-8:0]
root 9391 0.0 0.0 0 0 ? I< 12:42 0:00 [kworker/u518:2-ttm]
root 9392 0.0 0.0 0 0 ? I 12:42 0:00 [kworker/1:0]
root 9394 0.0 0.0 0 0 ? I 12:42 0:00 [kworker/0:1-events]
root 9395 0.0 0.0 0 0 ? I 12:42 0:00 [kworker/0:2]
root 9653 0.0 0.0 0 0 ? I 12:47 0:00 [kworker/u514:2-events_unbound]
atharva 9664 1.3 1.7 484620 30320 ? SNL 12:47 0:00 /usr/libexec/localsearch-extractor-3 --socket-fd 3
atharva 9676 6600 0.2 21180 4916 pts/1 R+ 12:47 0:00 ps aux
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ps aux > outputs/running_processes.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ |
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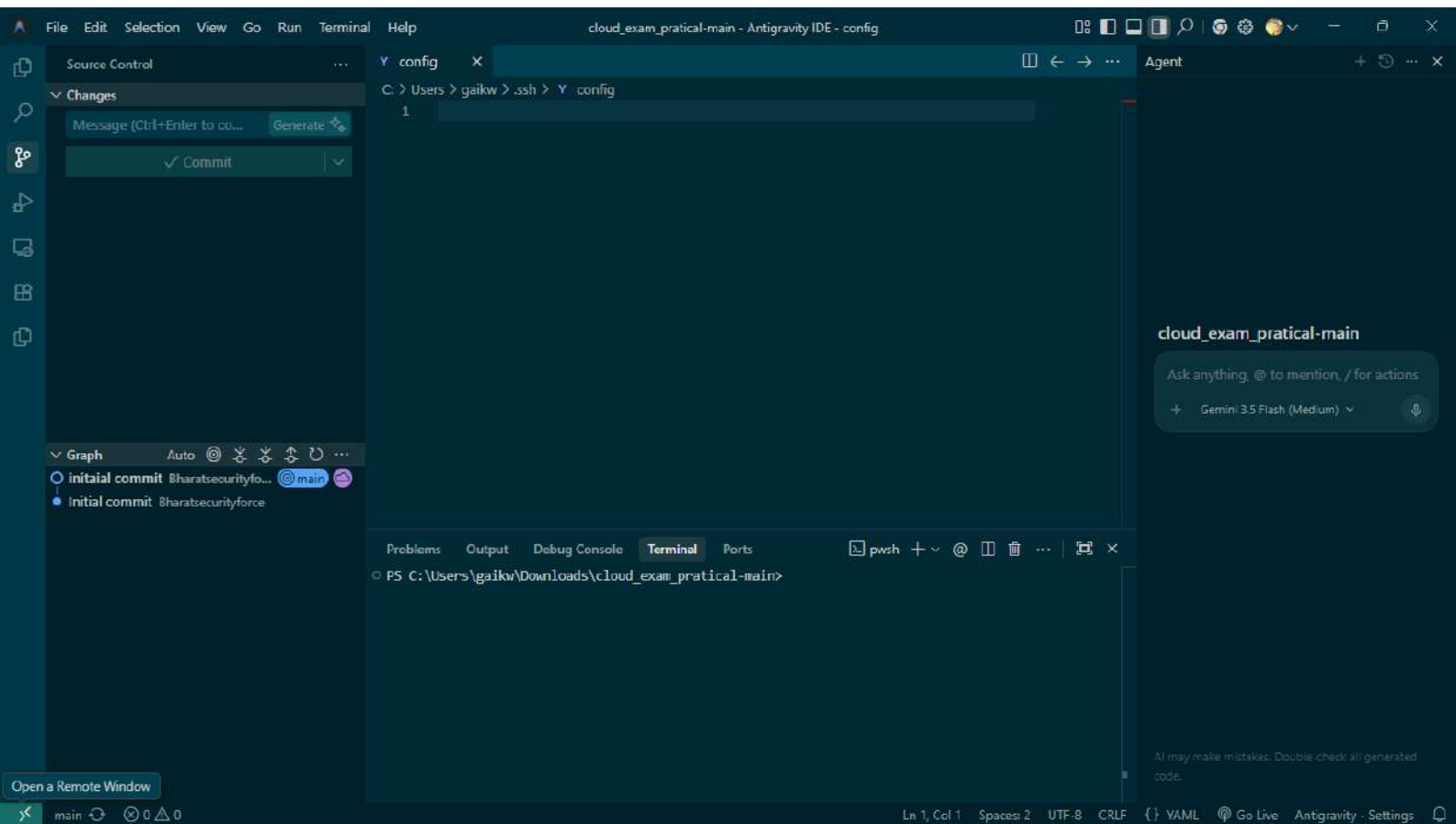
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ps aux > outputs/running_processes.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ find ~/practical_exam -name "*.sh"
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls scripts
file1.txt file2.txt file4.txt renamed_file3.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls outputs
disk_usage.txt file1.txt file2.txt memory_usage.txt running_processes.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ touch ~/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod 640 ~/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls -l ~/practical_exam/secure_data.txt
-rw-r----- 1 atharva atharva 0 Jun 10 12:50 /home/atharva/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls -l ~/practical_exam/secure_data.txt
-rw-r----- 1 atharva atharva 0 Jun 10 12:50 /home/atharva/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ id
uid=1000(atharva) gid=1000(atharva) groups=1000(atharva),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ id > ~/practical_exam/outputs/user_info.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ cat ~/practical_exam/outputs/user_info.txt
uid=1000(atharva) gid=1000(atharva) groups=1000(atharva),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/welcome.sh
Welcome to Cloud Computing
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/user_welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/user_welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/user_welcome.sh
Enter username:
atharva
Welcome atharva
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/larger.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/larger.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/larger.sh
Enter first number:
10
Enter second number:
01
10 is larger
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/even_odd.sh

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atharva@atharva-VMware-Vir X + v
disk_usage.txt file1.txt file2.txt memory_usage.txt running_processes.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ touch ~/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod 640 ~/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls -l ~/practical_exam/secure_data.txt
-rw-r----- 1 atharva atharva 0 Jun 10 12:50 /home/atharva/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls -l ~/practical_exam/secure_data.txt
-rw-r----- 1 atharva atharva 0 Jun 10 12:50 /home/atharva/practical_exam/secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ id
uid=1000(atharva) gid=1000(atharva) groups=1000(atharva),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ id > ~/practical_exam/outputs/user_info.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ cat ~/practical_exam/outputs/user_info.txt
uid=1000(atharva) gid=1000(atharva) groups=1000(atharva),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/welcome.sh
Welcome to Cloud Computing
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/user_welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/user_welcome.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/user_welcome.sh
Enter username:
atharva
Welcome atharva
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/larger.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/larger.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/larger.sh
Enter first number:
10
Enter second number:
01
10 is larger
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/even_odd.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/even_odd.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/even_odd.sh
Enter a number:
1001
Odd Number
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ |
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```
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/larger.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/larger.sh
Enter first number:
10
Enter second number:
01
10 is larger
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/even_odd.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/even_odd.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/even_odd.sh
Enter a number:
1001
Odd Number
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/for_loop.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/for_loop.sh
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/for_loop.sh
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ |
```

```
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ nano ~/practical_exam/scripts/while_loop.sh  
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ chmod +x ~/practical_exam/scripts/while_loop.sh  
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ~/practical_exam/scripts/while_loop.sh  
20  
19  
18  
17  
16  
15  
14  
13  
12  
11  
10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ |
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```
C:\WINDOWS\system32\cmd. X + | v
C:\Users\gaikw>ssh atharva@192.168.103.129
atharva@192.168.103.129's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jun 10 12:33:59 2026 from 192.168.103.1
atharva@atharva-VMware-Virtual-Platform:~$ nano ~/practical_exam/scripts/table.sh
atharva@atharva-VMware-Virtual-Platform:~$ chmod +x ~/practical_exam/scripts/table.sh
atharva@atharva-VMware-Virtual-Platform:~$ ~/practical_exam/scripts/table.sh
Enter a number:
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
atharva@atharva-VMware-Virtual-Platform:~$ nano ~/practical_exam/scripts/count_files.sh
atharva@atharva-VMware-Virtual-Platform:~$ chmod +x ~/practical_exam/scripts/count_files.sh
atharva@atharva-VMware-Virtual-Platform:~$ ~/practical_exam/scripts/count_files.sh
Total files: 12
```

```

C:\WINDOWS\system32\cmd. X + v
atharva@atharva-VMware-Virtual-Platform:~$ nano ~/practical_exam/scripts/list_scripts.sh
atharva@atharva-VMware-Virtual-Platform:~$ chmod +x ~/practical_exam/scripts/list_scripts.sh
atharva@atharva-VMware-Virtual-Platform:~$ ~/practical_exam/scripts/list_scripts.sh
/home/atharva/practical_exam/scripts/count_files.sh /home/atharva/practical_exam/scripts/table.sh
/home/atharva/practical_exam/scripts/even_odd.sh /home/atharva/practical_exam/scripts/user_welcome.sh
/home/atharva/practical_exam/scripts/for_loop.sh /home/atharva/practical_exam/scripts/welcome.sh
/home/atharva/practical_exam/scripts/larger.sh /home/atharva/practical_exam/scripts/while_loop.sh
/home/atharva/practical_exam/scripts/list_scripts.sh
atharva@atharva-VMware-Virtual-Platform:~$ nano ~/practical_exam/scripts/date_time.sh
atharva@atharva-VMware-Virtual-Platform:~$ chmod +x ~/practical_exam/scripts/date_time.sh
atharva@atharva-VMware-Virtual-Platform:~$ ~/practical_exam/scripts/date_time.sh > ~/practical_exam/outputs/date_time.txt
atharva@atharva-VMware-Virtual-Platform:~$ cat ~/practical_exam/outputs/date_time.txt
Wed Jun 10 05:32:10 PM IST 2026
atharva@atharva-VMware-Virtual-Platform:~$ ps aux > ~/practical_exam/outputs/processes.txt
atharva@atharva-VMware-Virtual-Platform:~$ head ~/practical_exam/outputs/processes.txt
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  2.1  0.9 26040 17036 ?        Ss   17:28   0:05 /usr/lib/systemd/systemd --switched-root --system --deserialize=51 auto automatic
-ubiquity noprompt
root         2  0.0  0.0      0     0 ?        S    17:28   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    17:28   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   17:28   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0      0     0 ?        I<   17:28   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0      0     0 ?        I<   17:28   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0      0     0 ?        I<   17:28   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   17:28   0:00 [kworker/R-netns]
root         9  0.0  0.0      0     0 ?        I    17:28   0:00 [kworker/0:0-events]
atharva@atharva-VMware-Virtual-Platform:~$ ps aux --sort=-%mem | head -6
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
gdm-gre+  2137  2.4 12.4 4768112 219996 ?        Ssl  17:28   0:06 /usr/bin/gnome-shell --mode=gdm
gdm-gre+  2459  0.2  2.5 772904 45764 ?        Ssl  17:28   0:00 /usr/libexec/xdg-desktop-portal-gnome
root      3282  7.6  2.5 392708 45332 ?        Ssl  17:32   0:01 /usr/libexec/fwupd/fwupd
root      1171  1.1  2.3 1998372 40972 ?        Ssl  17:28   0:03 /snap/snapd/current/usr/lib/snapd/snapd
gdm-gre+  2341  0.0  1.8 2746864 33192 ?        Sl   17:28   0:00 /usr/bin/gjs -m /usr/share/gnome-shell/org.gnome.ScreenSaver
atharva@atharva-VMware-Virtual-Platform:~$ ps aux --sort=-%mem | head -6 > ~/practical_exam/outputs/top5_memory_processes.txt
atharva@atharva-VMware-Virtual-Platform:~$ cat ~/practical_exam/outputs/top5_memory_processes.txt
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
gdm-gre+  2137  2.3 12.4 4768112 219996 ?        Ssl  17:28   0:06 /usr/bin/gnome-shell --mode=gdm

```

```
C:\WINDOWS\system32\cmd. X + | v
gdm-gre+ 2341 0.0 1.8 2746864 33192 ? Sl 17:28 0:00 /usr/bin/gjs -m /usr/share/gnome-shell/org.gnome.ScreenSaver
atharva@atharva-VMware-Virtual-Platform:~$ crontab -e
no crontab for atharva - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano          <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
atharva@atharva-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow command
* * * * * date >> /home/atharva/practical_exam/logs/cron_log.txt
atharva@atharva-VMware-Virtual-Platform:~$ cat ~/practical_exam/logs/cron_log.txt
Wed Jun 10 05:34:01 PM IST 2026
```

```
C:\WINDOWS\system32\cmd. X + v
# m h dom mon dow  command
* * * * * date >> /home/atharva/practical_exam/logs/cron_log.txt
atharva@atharva-VMware-Virtual-Platform:~$ cat ~/practical_exam/logs/cron_log.txt
Wed Jun 10 05:34:01 PM IST 2026
atharva@atharva-VMware-Virtual-Platform:~$ find ~/practical_exam -type f
/home/atharva/practical_exam/secure_data.txt
/home/atharva/practical_exam/logs/cron_log.txt
/home/atharva/practical_exam/scripts/file1.txt
/home/atharva/practical_exam/scripts/while_loop.sh
/home/atharva/practical_exam/scripts/for_loop.sh
/home/atharva/practical_exam/scripts/date_time.sh
/home/atharva/practical_exam/scripts/table.sh
/home/atharva/practical_exam/scripts/file2.txt
/home/atharva/practical_exam/scripts/welcome.sh
/home/atharva/practical_exam/scripts/larger.sh
/home/atharva/practical_exam/scripts/file4.txt
/home/atharva/practical_exam/scripts/count_files.sh
/home/atharva/practical_exam/scripts/user_welcome.sh
/home/atharva/practical_exam/scripts/renamed_file3.txt
/home/atharva/practical_exam/scripts/even_odd.sh
/home/atharva/practical_exam/scripts/list_scripts.sh
/home/atharva/practical_exam/outputs/file1.txt
/home/atharva/practical_exam/outputs/file2.txt
/home/atharva/practical_exam/outputs/date_time.txt
/home/atharva/practical_exam/outputs/memory_usage.txt
/home/atharva/practical_exam/outputs/top5_memory_processes.txt
/home/atharva/practical_exam/outputs/processes.txt
/home/atharva/practical_exam/outputs/disk_usage.txt
/home/atharva/practical_exam/outputs/running_processes.txt
/home/atharva/practical_exam/outputs/user_info.txt
atharva@atharva-VMware-Virtual-Platform:~$ cd ~/practical_exam
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ ls
logs  outputs  scripts  secure_data.txt
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ cd
atharva@atharva-VMware-Virtual-Platform:~$ ls
Desktop  Downloads  Pictures  Templates  cloud_course  condition.sh  first.sh  practical_exam
Documents  Music  Public  Videos  cloud_courses  conditions.sh  greetings.sh  snap
```

```
C:\WINDOWS\system32\cmd. X + v
atharva@atharva-VMware-Virtual-Platform:~$ cd ~/practical_exam
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ git init
Command 'git' not found, but can be installed with:
sudo apt install git
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [106 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [33.1 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [2,860 B]
Hit:7 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [211 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [113 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [35.4 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [2,988 B]
Fetched 780 kB in 2s (328 kB/s)
18 packages can be upgraded. Run 'apt list --upgradable' to see them.
atharva@atharva-VMware-Virtual-Platform:~/practical_exam$ sudo apt install git -y
Installing:
git

Installing dependencies:
git-man liberror-perl

Suggested packages:
git-doc git-email git-gui gitk gitweb git-cvs git-svn

Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 18
Download size: 5,453 kB
Space needed: 28.4 MB / 10.0 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 liberror-perl all 0.17030-1 [23.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 git-man all 1:2.53.0-1ubuntu1 [1,202 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 git amd64 1:2.53.0-1ubuntu1 [4,227 kB]
```

```
atharva@atharva-VMware-Vir X + | v
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.103.129' (ED25519) to the list of known hosts.
atharva@192.168.103.129's password:
Permission denied, please try again.
atharva@192.168.103.129's password:
Permission denied, please try again.
atharva@192.168.103.129's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

atharva@atharva-VMware-Virtual-Platform:~$
```

```
atharva@atharva-VMware-Vir X + | v
Permission denied, please try again.
atharva@192.168.103.129's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

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applicable law.

atharva@atharva-VMware-Virtual-Platform:~$ mkdir -p ~/practical_exam/scripts
atharva@atharva-VMware-Virtual-Platform:~$ mkdir -p ~/practical_exam/outputs
atharva@atharva-VMware-Virtual-Platform:~$ mkdir -p ~/practical_exam/logs
atharva@atharva-VMware-Virtual-Platform:~$ ls -R ~/practical_exam
/home/atharva/practical_exam:
logs  outputs  scripts

/home/atharva/practical_exam/logs:

/home/atharva/practical_exam/outputs:

/home/atharva/practical_exam/scripts:
atharva@atharva-VMware-Virtual-Platform:~$ |
```